Caitao Zhan

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About Me

I now work in `quantum` networking. Previously, I worked in `classical` networking and sensing.

Education

Aug. 2017 ~ Jan. 2024	Stony Brook University GPA: 3.9/4.0	Ph.D. Candidate in Computer Science Advisor: Himanshu Gupta, Thesis: PDF
Sept. 2013 ~ Jun. 2017	China University of Geosciences GPA: 92/100	B.S. in Computer Science and Technology Rank: 1/122

Employment

Feb. 2024 ~ Present	Postdoc @ Argonne National Lab (Data Science and Learning Division) Quantum networks, SeQUeNCe, Quantum Internet architecture
May 2021 ~ Aug. 2021	Software Engineering Intern @ Microsoft (Azure) `C#` development: Active Directory/Light-weight Directory Service (AD/LDS) Replay Tool Automation
Jun. 2018 ~ Jan. 2024	Research assistant @ Stony Brook University Wireless sensor networks, data center networks, quantum networks, and quantum sensor networks
Sep. 2017 ~ May 2018	Teaching assistant @ Stony Brook University

Research Experience

Feb. 2024 ~ Present	Quantum Networks and SeQUeNCe. [13, 14, 16, 17, 18, 19, 20, 21, 22, 23]
Jan. 2021 ~ Jan. 2024	Efficient Quantum Communication Networks. [8, 15] Design/implement routing algorithms/protocols for quantum networks. Bell states, GHZ, Graph states.
Sep. 2021 ~ Jan. 2024	Discrete Outcome Quantum Sensor Networks. [9, 12] Quantum state/channel discrimination, initial state optimization, semidefinite programming, theory.
Aug. 2022 ~ Sep. 2023	Quantum Sensor Network Algorithms for Transmitter Localization. [10] Quantum sensing, quantum state discrimination, quantum machine learning
Nov. 2019 ~ Mar. 2022	Intelligent Radio with Deep Learning. [5, 7, 11] Design/implement CNNs to solve wireless network problems: wireless localization & spectrum allocation. Reframe wireless problems to computer vision problems: image-to-image translation & object detection.
Mar. 2019 ~ Oct. 2019	Efficient Localization of Multiple Intruders in Shared Spectrum System. [3] Design/implement. Bayesian approach. Testbed(Odroid,Raspberry Pi,USRP,HackRF).
Dec 2018 ~ Sep. 2020	Datacenter Networks. [4] Multi-hop circuit switch scheduling. Greedy, approximation proof. Participate in implementation.
July 2018 ~ July 2019	Selection of Sensors for Efficient Transmitter Localization. [2, 6] Implement. Greedy, approximation proof. Bayesian approach. GPU acceleration.
Oct. 2015 ~ Sept. 2016	Optimization using Evolutionary Algorithms. [1] Design/implement. Shortest path-finding using ant colony optimization algorithms. Proposed a probability-based evolutionary algorithm solving shape formation problems.

Selected Awards and Honors

China National Scholarship	2014, Chinese Ministry of Education, Top 1%
Travel Grant for ACM IMC	2018, ACM Internet Measurement Conference
Best Poster Award (Graduate Research Day)	2022, Department of CS, Stony Brook University
Outstanding Poster Presentation Award (PRCS 2024)	2024, Argonne National Laboartory
Best Paper Award (QNET track)	2025, IEEE Quantum Week (QCE)
Postdoctoral Performance Award (PRCS 2025 Co-chair)	2025, Argonne National Laboratory
LDRD Seed Award (Principal Investigator)	2026, Argonne National Laboratory, $25K
R&D 100 Award (SeQUeNCe)	2026, R&D World

Academic Services

`Co-chair` of Argonne National Lab Postdoctoral Research and Career Symposium (PRCS) 2025
`Program Committee` of conferences IEEE QCE 2024 ∼ 26, QCNC 2025 ∼ 27, ICCCN 2026, and many workshops
`Reviewer`: Elsevier J. Netw. Comput. Appl., Elsevier PMC, IEEE TON, IEEE IoT-J, IEEE OJCOMS, IEEE TCOM, ACM TQC, Springer Nature npj Quantum Inf., IEEE TQE, ISIT 2025, Euro-Par 2025.

Papers

[23] Sanjay Suresh, **Caitao Zhan**, Joaquin Chung, Rajkumar Kettimuthu, "SeQUeNCe.DQC: Full-stack Simulation of Distributed Quantum Computing", arXiv

[22] Joaquin Chung, Michal Hajdušek, ···, **Caitao Zhan**, Raj Kettimuthu, and Rodney Van Meter, "Cross-Validation of Open-Source Quantum Network Simulators", arXiv

[21] Peter Hoang, Phani R V Marthi, **Caitao Zhan**, Rajkumar Kettimuthu, Sagar Patange, Suman Debnath, "End-to-End Connectivity of Quantum Resources with EMT Power Grid Simulator", CIGRE GOTF 2026, PDF

[20] Sagar Patange, **Caitao Zhan**, Bikun Li, Joaquin Chung, Allen Zang, Liang Jiang, Rajkumar Kettimuthu, "Realistic Simulation of Quantum Repeater with Encoding and Classical Error Correction", IEEE QCE 2026, arXiv

[19] Joaquin Chung, Daniel Dilley, ···, **Caitao Zhan**, et al., "InterQnet: A Heterogeneous Full-Stack Approach to Co-designing Scalable Quantum Networks", IEEE Transactions on Quantum Engineering 2026, arXiv

[18] Hayden Miller, **Caitao Zhan**, Michael Bishof, Joaquin Chung, Han Xu, Prem Kumar, Rajkumar Kettimuthu, "Simulation of a Heterogeneous Quantum Network", QCNC 2026, arXiv

[17] Xinan Chen, **Caitao Zhan**, Joaquin Chung and Jeffrey Larson, "Routing Entanglement in Complex Quantum Networks Using GHZ States", QCNC 2026, arXiv

[16] Laura d'Avossa, **Caitao Zhan**, Joaquin Chung, Rajkumar Kettimuthu, Angela Sara Cacciapuoti, Marcello Caleffi, "Simulation of Quantum Transduction Strategies for Quantum Networks", IEEE QCE 2025, PDF. *Best Paper Award.*

[15] X. Fan, **C. Zhan**, H. Gupta, C.R. Ramakrishnan, "Optimized Distribution of Entanglement Graph States in Quantum Networks", IEEE Transactions on Quantum Engineering, 2025, PDF.

[14] **Caitao Zhan**, Joaquin Chung, Rajkumar Kettimuthu, Allen Zang, Alexander Kolar, "Design and Simulation of the Adaptive Continuous Entanglement Generation Protocol", QCNC 2025, arXiv.

[13] Francesco Mazza, **Caitao Zhan**, Joaquin Chung, Rajkumar Kettimuthu, Marcello Caleffi, Angela Sara Cacciapuoti, "Simulation of Entanglement-Enabled Connectivity in QLANs using SeQUeNCe", IEEE ICC 2025, arXiv.

[12] **C. Zhan**, H. Gupta, M. Hillery, "Optimizing Initial State of Detector Sensors in Quantum Sensor Networks", ACM Transactions on Quantum Computing, 2024. PDF.

[11] M. Ghaderibaneh, **C. Zhan**, H. Gupta, "DeepAlloc: CNN-Based Approach to Efficient Spectrum Allocation in Shared Spectrum Systems", IEEE Access 2024, PDF.

[10] **C. Zhan**, H. Gupta, "Quantum Sensor Network Algorithms for Transmitter Localization", IEEE QCE 2023, PDF.

[9] M. Hillery, H. Gupta, **C. Zhan**, "Discrete Outcome Quantum Sensor Networks", Physical Review A, 2023, PDF.

[8] M. Ghaderibaneh, **C. Zhan**, C.R. Ramakrishnan, H. Gupta, "Efficient Quantum Network Communication using Optimized Entanglement-Swapping Trees", IEEE Transactions on Quantum Engineering, 2022. PDF.

[7] **C. Zhan**, M. Ghaderibaneh, P. Sahu, H. Gupta, "DeepMTL Pro: Deep Learning Based Multiple Transmitter Localization and Power Estimation", Elsevier Pervasive and Mobile Computing, 2022, PDF.

[6] A. Bhattacharya, **C. Zhan**, A. Maji, H. Gupta, S. Das, P. Djuric, "Selection of Sensors for Efficient Transmitter Localization", IEEE/ACM Transactions on Networking 2021, PDF.

[5] **C. Zhan**, M. Ghaderibaneh, P. Sahu, H. Gupta, "DeepMTL: Deep Learning Based Multiple Transmitter Localization", IEEE WoWMoM 2021, PDF.

[4] H. Gupta, M. Curran, **C. Zhan**, "Near-Optimal Multihop Scheduling in General Circuit-Switched Networks", ACM CoNEXT 2020, PDF.

[3] **C. Zhan**, H. Gupta, A. Bhattacharya, M. Ghaderibaneh, "Efficient Localization of Multiple Intruders in Shared Spectrum System", ACM/IEEE IPSN 2020, PDF.

[2] A. Bhattacharya, **C. Zhan**, H. Gupta, S. Das, P. Djuric, "Selection of Sensors for Efficient Transmitter Localization", IEEE INFOCOM 2020, PDF.

[1] **C. Zhan** and C. Li, "Shape Formation in Games: a Probability-based Evolutionary Approach", 2016 International Conference on Computational Intelligence and Security, PDF.